ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** less than 1 per cent of the radius	35/6400 is about 0.5 per cent.
2	**(c)** 105 L	21/100 x 500.
3	**(c)** 3 per cent	Most is salty.
4	**(c)** a thick CO2 atmosphere	Runaway greenhouse.
5	**(a)** extremely high pressure	Pressure keeps it solid.
6	**(c)** 156 L	78/100 x 200.
7	**(c)** 30	3 per cent.
8	**(d)** is farther from the Sun and has a thin atmosphere	Weak greenhouse.
9	**(c)** seismic waves of earthquakes	Waves travel through layers.
10	**(b)** air is thinner with lower oxygen amount per breath	Pressure falls.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The pressure is enormous (1); It keeps the iron-nickel solid (1).	1+1
12	Troposphere is heated from the ground (1); Stratosphere is heated by ozone absorbing UV (1).	1+1
13	Air is thin at height (1); Less oxygen in each breath (1).	1+1
14	21/100 x 250 (1); = 52.5 L (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius = crust + mantle + core (1); Core = 6400 - 35 - 2900 (1); = 3465 km (1).	1+1+1
16	Earth has suitable temperature and air pressure (1); Moon has no atmosphere (1); Water would evaporate or freeze (1).	1+1+1
17	Air molecules scatter blue light more (1); Blue reaches our eyes from all directions (1); In space the sky is black (1).	1+1+1
	OR: Earth was once molten (1); Dense iron sank to the centre (1); Lighter rocks rose to the crust (1).	1+1+1
18	Floods then drying of rivers (1); Sea level rises (1); Coastal areas submerged (1).	1+1+1
	OR: Oceans dissolve CO2 (1); Release it when warm (1); Marine plants use it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Crust: 5 to 70 km, solid rock (1); Mantle: about 2900 km, hot semi-solid rock (1); Outer core: liquid iron-nickel (1); Inner core: solid iron-nickel (1); Seismic waves reveal it (1).	1+1+1+1+1
	OR: CFCs from refrigerators (1); Release chlorine in the stratosphere (1); UV reaches Earth (1); Montreal Protocol (1); Layer slowly recovering (1).	1+1+1+1+1
20	Troposphere (to about 12 km): weather (1); Stratosphere (to about 50 km): ozone (1); Mesosphere (to about 85 km): meteors burn (1); Thermosphere: auroras and satellites (1); Exosphere: merges with space (1).	1+1+1+1+1
	OR: Atmosphere (1); Temperature (1); Water (1); Gravity (1); Life (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** fossil groundwater in deep layers	Replenished very slowly.
2	**(c)** has no atmosphere	No insulation.
3	**(b)** mantle	About 2900 km.
4	**(a)** ozone absorbs UV radiation	Heating by ozone.
5	**(a)** more is drawn than is recharged	Overuse.
6	**(d)** diamonds	Not required.
7	**(c)** thousands of degrees Celsius	Very hot.
8	**(d)** the outside pressure is low and the gas inside expands	Pressure falls.
9	**(d)** groundwater level	Recharging.
10	**(a)** 150 million km	1 AU.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Waves change speed and direction in different layers (1); Recorded at stations show the layered structure (1).	1+1
12	Traps heat at night (1); Reduces the heat received during the day (1).	1+1
13	In the stratosphere it absorbs UV (1); At ground level ozone is a pollutant (1).	1+1
14	Traps heat (1); Makes the surface very hot (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Troposphere: weather, temperature falls (1); Stratosphere: ozone, temperature rises (1); Troposphere holds most air (1).	1+1+1
16	Removal of salts from sea water (1); By distillation or reverse osmosis (1); Gives drinking water in dry areas (1).	1+1+1
17	Remove vegetation (1); Cause erosion and pollution (1); Destroy habitats (1).	1+1+1
	OR: Heat and pressure in the interior drive plate motion (1); Volcanoes release magma (1); Earthquakes release stored energy (1).	1+1+1
18	Use less electricity (1); Use public transport or cycle (1); Plant trees (1).	1+1+1
	OR: Gravity: Moon about one-sixth (1); Atmosphere: Earth has, Moon none (1); Water: Earth has liquid water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Natural greenhouse effect keeps Earth warm (1); Extra CO2 from fuels enhances it (1); Venus: strong greenhouse (1); Mars: weak greenhouse (1); Earth is in between (1).	1+1+1+1+1
	OR: Air: vehicles and industries (1); Water: sewage and plastics (1); Effects: disease and death of life (1); Control: filters, treatment (1); Public action (1).	1+1+1+1+1
20	Liquid water (1); Oxygen-rich air (1); Distance from the Sun (1); Magnetic field and ozone (1); Other bodies lack some conditions (1).	1+1+1+1+1
	OR: Ice cores trap ancient air (1); Tree rings show wet and dry years (1); Fossils show past life (1); Reveal temperature trends (1); Help predict future changes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** crust, mantle, outer core, inner core	Surface to centre.
2	**(a)** It contains most of the water vapour and clouds	Weather layer.
3	**(a)** availability of safe water	Contamination.
4	**(b)** 8 minutes	About 500 s.
5	**(a)** mantle (magma)	Molten rock.
6	**(d)** Antarctica	Seasonal depletion.
7	**(d)** desalination	Remove salts.
8	**(d)** motion of the liquid outer core	Dynamo effect.
9	**(c)** sudden release of energy in the crust	Movement of plates.
10	**(a)** avoiding products with CFCs	CFC-free appliances.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CFCs destroy ozone (1); Protects the UV shield (1).	1+1
12	Air is calm with little weather (1); Fewer clouds and storms (1).	1+1
13	Only about 3 per cent of Earth's water is fresh (1); Most of it is frozen or underground (1).	1+1
14	They take in CO2 and release oxygen (1); Regulate air quality (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Deflects solar wind (1); Protects the atmosphere (1); Reduces radiation (1).	1+1+1
16	Gravity pulls gases down (1); Density is highest near the ground (1); Thins out with height (1).	1+1+1
17	Plants use light for photosynthesis (1); Sun's heat drives the water cycle (1); Drives winds (1).	1+1+1
	OR: Ozone absorbs UV (1); CFCs cause depletion (1); Protocol banned CFCs (1).	1+1+1
18	Tilt of 23.5 degrees (1); Sun's rays fall differently (1); Changing seasons affect crops and animals (1).	1+1+1
	OR: Crop yield may fall (1); More evaporation, drought (1); Floods due to heavy rains (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	97 per cent salt, 3 per cent fresh (1); Most fresh water is frozen (1); Rainwater harvesting (1); Efficient irrigation (1); Stop pollution and reuse water (1).	1+1+1+1+1
	OR: Solar and wind energy (1); Reduce, reuse, recycle (1); Afforestation (1); Water conservation (1); Responsible consumption (1).	1+1+1+1+1
20	Rain: hydrosphere and atmosphere (1); Soil: weathering of lithosphere plus organisms (1); Forest: plants use water, soil and air (1); Biosphere depends on all (1); Changes in one affect the others (1).	1+1+1+1+1
	OR: Rotation: day-night (1); Revolution: year (1); Tilt: seasons (1); Organisms adapt to cycles (1); Moderate temperature (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.